MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

SEC- MG5SECECT302

LOW VOLTAGE SYSTEM FOR BUILDING MANGEMENT

PART A

1. [a] consumes a significant portion of building energy	[U] [CO1]
2. [d] all of the above	[U] [CO1]
3. [a] environmental sustainability	[U] [CO1]
4. [a] monitor, manage and optimize building services	[U] [CO1]
5. [c] reduce artificial lighting	[U][CO 1]
6. [a] field level	[U][CO 1]
7.[a] Servers and other BMS devices	[U][CO 1]
8.[a] chiller failure	[U][CO 1]
9.[d] all of these	[U][CO 1]
10. [a] turning off unnecessary lights	[U] [CO 1]
11.[a]provide clear emergency instructions	[U][CO1]
12.[b] authenticate users	[U][CO 1]
13. [b] controlling the conduction period of the AC waveform.	[A][CO2]
14.[a] measuring a wide range of temperatures	[A][CO2]
15.[c] Both A and B	[A][CO2]
16. [b] they provide faster switching, silent operation and longer service life.	[A][CO2]
17. [c] Both A and B	[A][CO2]
18. [d] electromagnetic interference (EMI)	[A][CO2]
19. [d] temperature fluctuations	[A][CO2]
20. [d] all of the above	[A][CO2]
21.[b] generate heat during cold weather	[A][CO2]

22. [a] heat transfer [A][CO2]

23. [a] at or near its full rated brightness [A][CO2]

24.[c] whether the elevator doors are open or closed [A][CO2]

25. [a] adjust the fan motor speed [A][CO2]

26. [d] all of the above [A][CO2]

27. [d] all of the above [A][CO2]

28.[c] above 25^0C [A][CO2]

29.[c] human movement [An][CO3]

30. [d]identify or authenticate a user [An][CO3]

31. [d] it can provide detailed information about device status and fault [An][CO3]

32.[a] unnecessary operation of equipment [A][CO4]

33.[b] detect, control, contain and respond to fire [A][CO4]

34. [b] allow a person to manually activate the fire alarm [A][CO4]

35. [d] all of the above [A][CO2]

PART B

Answer any 3 questions. Each question carries 5 marks

36. CCTV stands for Closed-Circuit Television. It is a video surveillance system used to monitor and record activities in a building. It consists of cameras, cables or network connections, a monitor, and a recording device (DVR/NVR). It is used for security, safety, and crime prevention. In Building Management System (BMS), CCTV helps monitor entrances, exits, corridors, parking areas, and other important locations. [U][CO1]

37. A TRIAC (Triode for Alternating Current) is a semiconductor switching device used to control AC (alternating current) power. It acts like an electronically controlled switch, allowing current to flow through a load in both directions when triggered. TRIACs have no mechanical contacts, reducing wear and tear. They switch electrical loads faster than mechanical relays. They operate without clicking or switching noise. They have a longer operating life due to the absence of contact wear. They require less maintenance and are suitable for frequent switching in building automation systems. [A][CO2]

38. Intrusion Detection System is a system that detects unauthorized entry or suspicious activity within a building and sends alerts to security personnel or a central monitoring system.
Detects unauthorized entry into buildings or restricted areas.
Uses door contacts, motion detectors, and glass-break sensors to detect intrusions.
Triggers alarms when suspicious activity or unauthorized access is detected.

Sends alerts to security personnel or the Building Management System (BMS).
Helps prevent theft, vandalism, and unauthorized access by enabling a quick response.

[A] [CO3]

39. Variable Frequency Drive is an electronic device used to control the speed of an AC motor by varying the frequency and voltage of the electrical supply. Controllers use VFDs to vary the motor speed by changing the supply frequency. Sensors measure parameters such as pressure, temperature, and airflow. {Use feedback path to control}. The controller automatically adjusts the VFD speed according to the required pressure, flow, or temperature. VFDs reduce motor speed when full capacity is not required, saving energy. By using VFDs they provide smooth starting and stopping, reduce mechanical stress, and improve system efficiency. [U][CO1]

40. DCV stands for Demand-Controlled Ventilation. It is a ventilation control strategy that automatically adjusts the amount of fresh outdoor air supplied to a building according to actual occupancy and ventilation demand. DCV helps maintain indoor air quality while reducing unnecessary ventilation, fan energy consumption, and heating or cooling loads. Reduces energy consumption by supplying ventilation only when required. Uses CO_2 sensors to adjust fresh air supply according to occupancy there by ensuring improved air quality. Reduces HVAC operating costs and electricity bills. Automatically adjusts ventilation based on the number of people in a room {occupancy-based control}. Used in offices, classrooms, shopping malls, conference rooms, and other modern buildings through the Building Management System (BMS). [A][CO2]